Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The co-ordinates of a moving particle at any time ' t 'are given by $x=\alpha t^3$ and $y=\beta t^3$. The speed of the particle at time ' t ' is given by

(A) $3t\sqrt{\alpha^2+\beta^2}$

(B) $3t^2\sqrt{\alpha^2+\beta^2}$

(C) $t^2\sqrt{\alpha^2+\beta^2}$

(D) $\sqrt{\alpha^2+\beta^2}$

Q.2

It is given that $\mathrm{t}=\mathrm{p}x^2+\mathrm{q}x$, where x is displacement and t is time. The acceleration of particle at origin is

(A) $-\frac{2p}{q^3}$

(B) $-\frac{2q}{p^3}$

(C) $\frac{2p}{q^3}$

(D) $\frac{2q}{p^3}$

Q.3

A particle moving along x -axis has acceleration f, at time t,
given by $\mathrm{f}=\mathrm{f}_0\left(1-\frac{\mathrm{t}}{\mathrm{T}}\right)$, where f_0 and T are constants. The particle at $\mathrm{t}=0$ has zero velocity. In the time interval between $t=0$ and the instant when $\mathrm{f}=0$, the particle's velocity $\left(\mathrm{v}_{\mathrm{x}}\right)$ is

(A) $\frac{1}{2}\mathrm{f}_0\ \mathrm{T}^2$

(B) $\mathrm{f}_0\ \mathrm{T}^2$

(C) $\frac{1}{2}\mathrm{f}_0\ \mathrm{T}$

(D) $\mathrm{f}_0\ \mathrm{T}$

Q.4

The displacement ' x ' (in meter) of a particle of mass ' m ' (in kg) moving in one dimension under the action of a force, is related to time ' t ' (in sec) by $t=\sqrt{x}+3$. The displacement of the particle when its velocity is zero, will be

(A) $2\ \mathrm{m}$

(B) $4\ \mathrm{m}$

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(C) zero

(D) $6\ \mathrm{m}$

Q.5

The dependence of velocity of a body with time is given by the equation $\mathrm{v} = 20 + 0.1\mathrm{t}^2$. The body is in

(A) uniform retardation

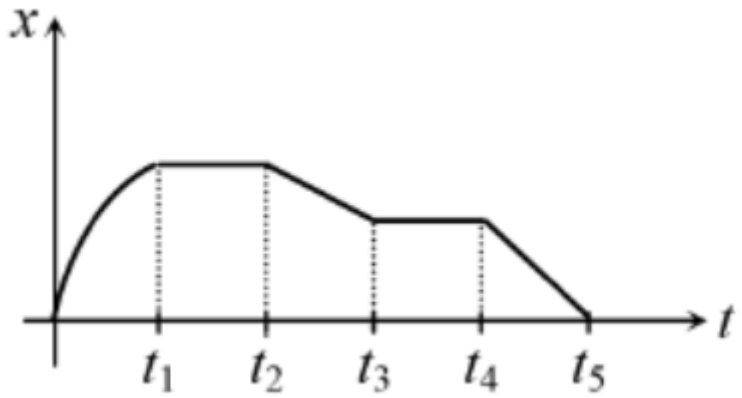
(B) uniform acceleration

(C) non-uniform acceleration

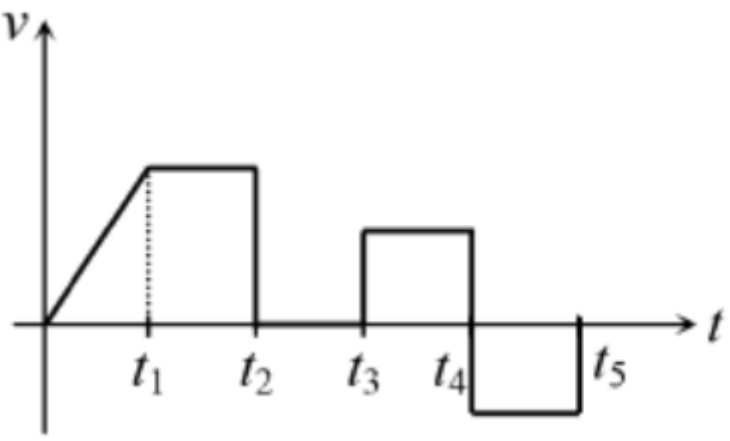
(D) zero acceleration.

Q.6

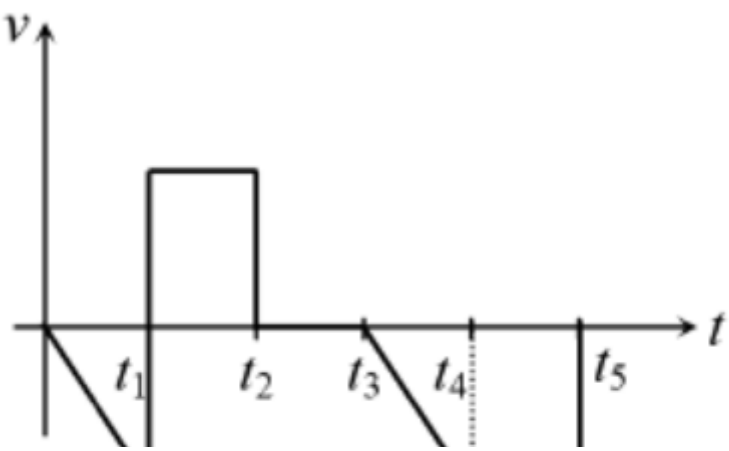
For position-time $(x - t)$ curve as shown in figure, the velocity-time $(v - t)$ curve will be



(A)

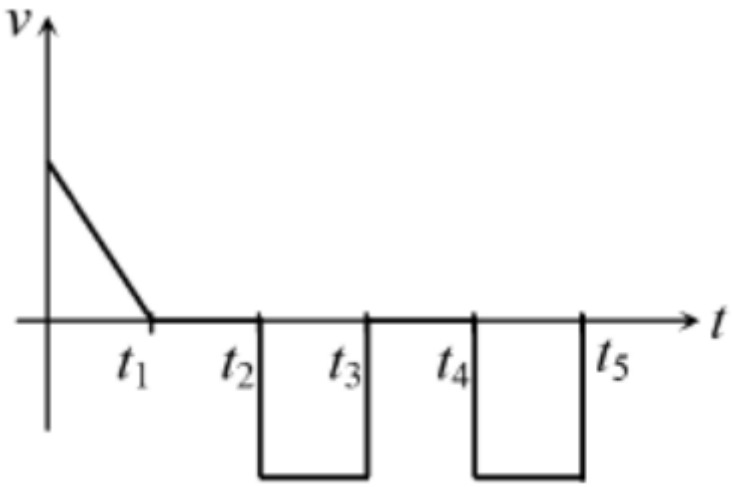


(B)

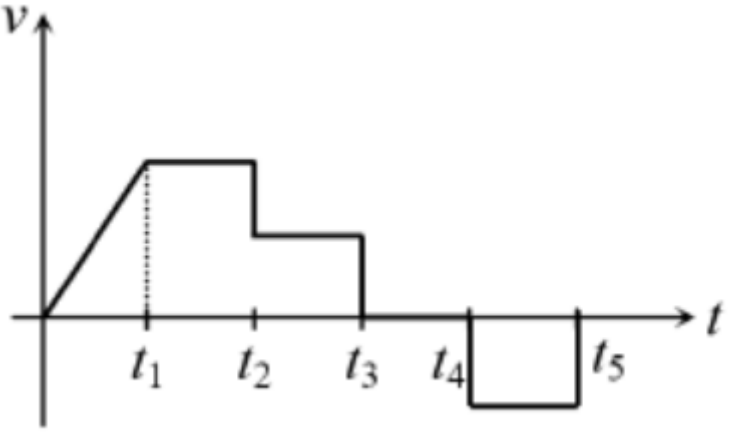


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(C)



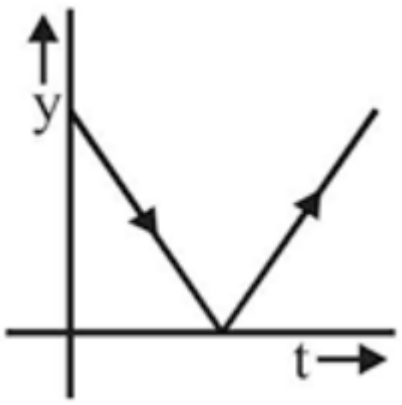
(D)



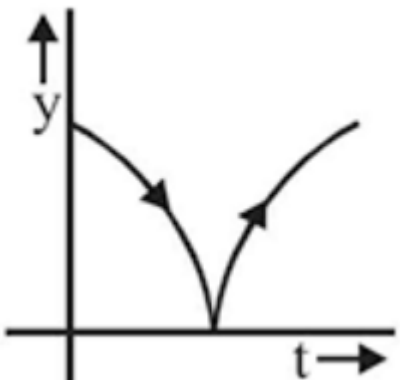
Q.7

A ball is dropped on a floor and bounces back to a height somewhat less than the original height. Which of the curves depicts its motion correctly?

(A)

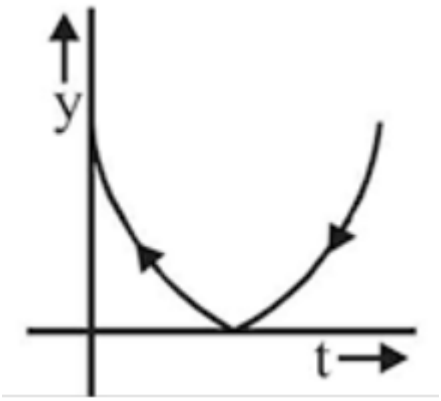


(B)

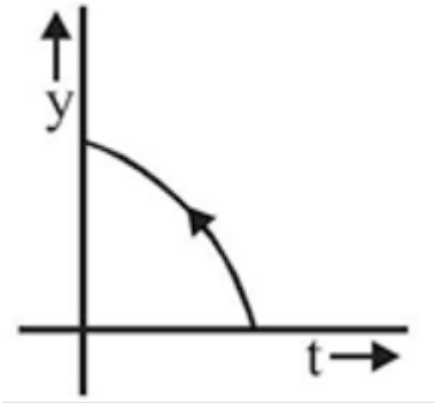


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(C)

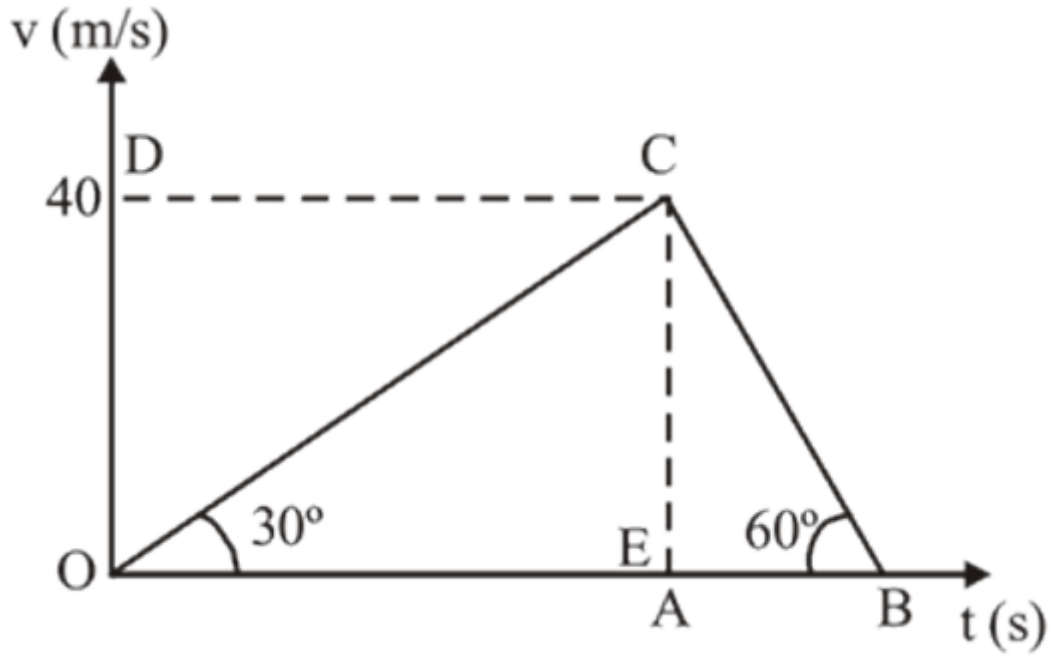


(D)



Q.8

The velocity-time graph of a body is shown in fig. The ratio of average acceleration during the intervals OA and AB is



(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) 3

Q.9

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The displacement-time graphs of two particles A and B are straight lines making angles of 30° and 60° respectively with the time axis. If the velocity of A is v_A and that of B is v_B, the value of v_A/v_B is

(A) $1/2$

(B) $1/\sqrt{3}$

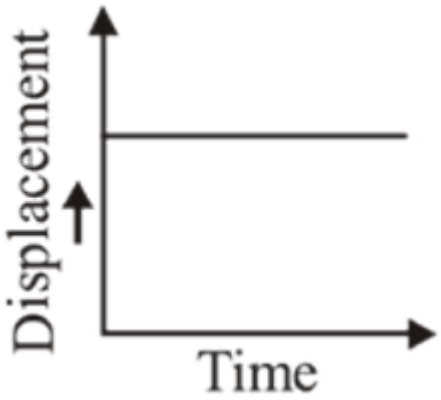
(C) $\sqrt{3}$

(D) $1/3$

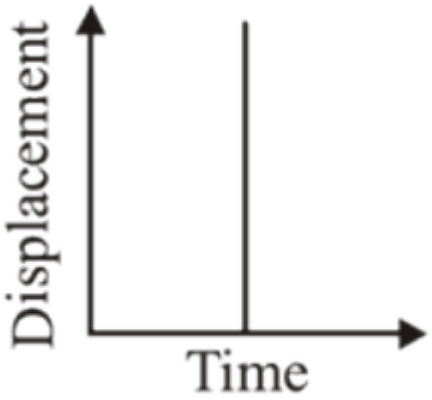
Q.10

Which of the following is not possible for a body in uniform motion?

(A)



(B)



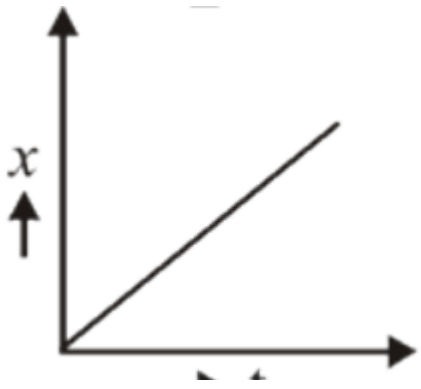
(C) Both a & b

(D) None of these

Q.11

Which of the following graphs gives the equation $x = v_o t + \frac{1}{2}at^2$

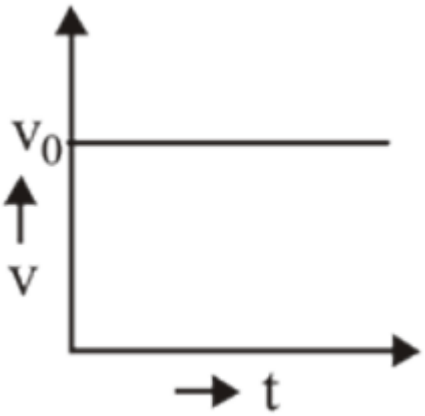
(A)



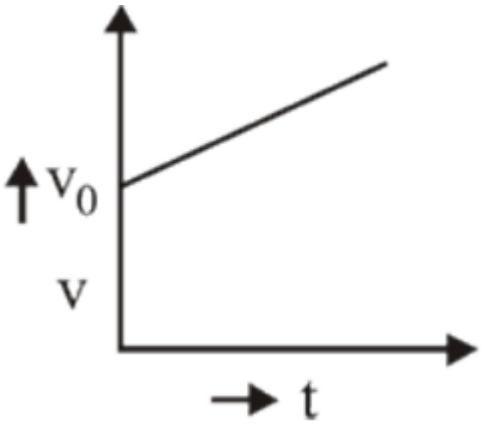
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→ t

(B)



(C)



(D) None of these

Q.12

A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β and comes to rest. If the total time elapsed is t, then the maximum velocity acquired by the car is

(A) $\left(\frac{\alpha^2+\beta^2}{\alpha\beta}\right) t$

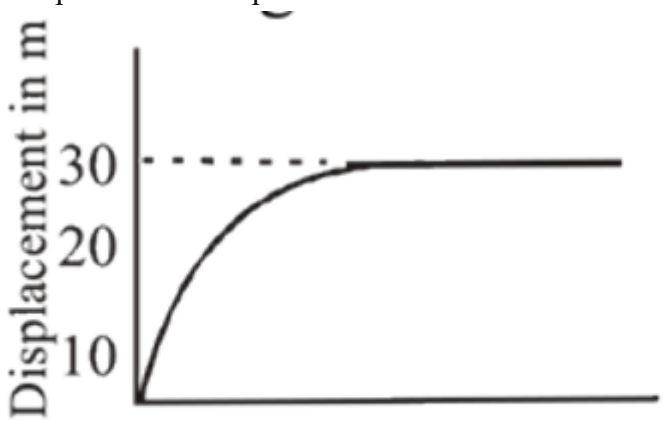
(B) $\left(\frac{\alpha^2-\beta^2}{\alpha\beta}\right)$

(C) $\frac{(\alpha+\beta)t}{\alpha\beta}$

(D) $\frac{\alpha\beta t}{\alpha+\beta}$

Q.13

The displacement of a particle as a function of time is shown in figure. It indicates that



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10 20 30 40 50

Time in sec

(A) the velocity of the particle is constant throughout

(B) the acceleration of the particle is constant throughout

(C) the particle starts with a constant velocity and is accelerated

(D) the motion is retarded and finally the particle stops

Q.14

A particle starts moving rectilinearly at time $t = 0$ such that its velocity v changes with time t according to equation $v = t^2 - t$ where t is in seconds and v in ms^{-1}. Find the time interval for which the particle retards.

(A) 0.5s < t < 1 s

(B) 0.5s < t < 2 s

(C) 0.5s < t < 3 s

(D) 0.5s < t < 5 s

Q.15

Consider a particle initially moving with a velocity of $5\ \mathrm{m\ s}^{-1}$ starts decelerating at a constant rate of $2\ \mathrm{m\ s}^{-2}$.
(a) Determine the time at which the particle becomes stationary.
(b) Find the distance travelled in the second second.
(c) Find the distance travelled in the third second.

(A) 2.5 s, 3 m, 0.5 m

(B) 2.5 s, 2 m, 5 m

(C) 5 s, 2 m, 0.5 m

(D) 2.5 s, 2 m, 0.5 m

Q.16

A car moves in a straight line, the car accelerates from rest with a constant acceleration α on a straight road. After gaining a velocity v, the car moves with that velocity for sometime. Then the car decelerates with a retardation β. If the total distance covered by the car is equal to s, find the total time of its motion.

(A) $\frac{s}{v} + v \times \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

(B) $s + \frac{v}{2}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

(C) $\frac{s}{v} + \frac{s}{2}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

(D) $\frac{s}{v} + \frac{v}{2}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

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Q.17

A car starts moving rectilinearly, first with acceleration $\alpha = 5\ \mathrm{ms}^{-2}$ (the initial velocity is equal to zero), then uniformly, and finally, decelerating at the same rate α comes to a stop. The total time of motion equals $t = 25\ \mathrm{s}$. The average velocity during this time is equal to $< y >= 72\mathrm{kmh}^{-1}$. How long does the car move uniformly?

(A) 15s

(B) 3s

(C) 4s

(D) 12s

Q.18

A particle retards from a velocity v_0 while moving in a straight line. If the magnitude of deceleration is directly proportional to the square root of the speed of the particle, find its average velocity for the total time of its motion.

(A) $\frac{v_0}{3}$

(B) $3v_0$

(C) $\frac{4v_0}{3}$

(D) $5v_0$

Q.19

Consider a solid cylinder of height h, density ρ_s, cross-sectional area A is floating in a liquid of density ρ_l. The frequency of vertical oscillations was found to be $\mathrm{g}^{\mathrm{a}}\mathrm{h}^{\mathrm{b}}\rho_i^{\mathrm{c}},\rho_{\mathrm{s}}^{\mathrm{d}}$. when depressed slightly and released. Then what is the value of $\frac{\mathrm{a}^2\mathrm{b}}{\mathrm{cd}}$?

(A) 0.4

(B) 0.6

(C) 0.3

(D) 0.5

Q.20

A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass $\mathrm{m}(\mathrm{m} << \mathrm{M})$. The speed of transverse waves in the string is $50\mathrm{ms}^{-1}$ when the car is at rest. But if the car is imparted an acceleration of a, the wave speed increases to $51\mathrm{ms}^{-1}$. What is the value of a? [Take $g = 10\mathrm{ms}^{-2}$

(A) 02.80

(B) 02.85

(C) 02.75

(D) 02.82

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Answers & Solutions

Q.1 Answer:

$3t^2\sqrt{\alpha^2+\beta^2}$

Solution:

$$x = \alpha t^3 \text{ and } y = \beta t^3$$
$$v_x = \frac{dx}{dt} = 3\alpha t^2 \text{ and } v_y = \frac{dy}{dt} = 3\beta t^2$$
$$\therefore v = \sqrt{v_x^2 + v_y^2} = \sqrt{9\alpha^2 t^4 + 9\beta^2 t^4}$$
$$= 3t^2\sqrt{\alpha^2+\beta^2}$$

Q.2 Answer:

$-\frac{2p}{q^3}$

Solution:

$t = px^2 + qx$

Differentiate with respect to time,

$1 = 2px\frac{dx}{dt} + q\frac{dx}{dt}$

or $2pxv + qv = 1$

At $x = 0, qv = 1$ **or** $v = \frac{1}{q}$

Again differentiate with respect to time,

Hence, $2pxa + 2pv^2 + qa = 0$

At $x = 0$**, put** $v = \frac{1}{q}$

Thus, we get $a = \frac{-2p}{q^3}$

Q.3 Answer:

$\frac{1}{2}f_0\ T$

Solution:

Here, $f = f_0\left(1 - \frac{t}{T}\right)$ **or,** $\frac{dv}{dt} = f_0\left(1 - \frac{t}{T}\right)$

or, $dv = f_0\left(1 - \frac{t}{T}\right)dt$

$\therefore v = \int dv = \int \left[f_0\left(1 - \frac{t}{T}\right)\right]dt$

or, $v = f_0\left(t - \frac{t^2}{2T}\right) + C$

where C is the constant of integration. At $t = 0, v = 0. \therefore 0 = f_0\left(0 - \frac{0}{2T}\right) + C \Rightarrow C = 0$

$\therefore v = f_0\left(t - \frac{t^2}{2T}\right)$

If $f = 0$**, then**

$0 = f_0\left(1 - \frac{t}{T}\right) \Rightarrow t = T$

Hence, particle's velocity in the time interval $t = 0$ **and** $t = T$ **is given by**

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$$v_x = \int_{t=0}^{t=T} dv = \int_{t=0}^{T} \left[f_0 \left(1 - \frac{t}{T} \right) \right] dt$$
$$= f_0 \left[\left(t - \frac{t^2}{2T} \right) \right]_0^T$$
$$= f_0 \left(T - \frac{T^2}{2T} \right) = f_0 \left(T - \frac{T}{2} \right)$$
$$= \frac{1}{2} f_0 T$$

Q.4 Answer:

zero

Solution:

$\because t = \sqrt{x} + 3$

$\Rightarrow \sqrt{x} = t - 3 \Rightarrow x = (t-3)^2$

$v = \dfrac{dx}{dt} = 2(t-3) = 0$

$\Rightarrow t = 3$

$\therefore x = (3-3)^2$

$\Rightarrow x = 0$

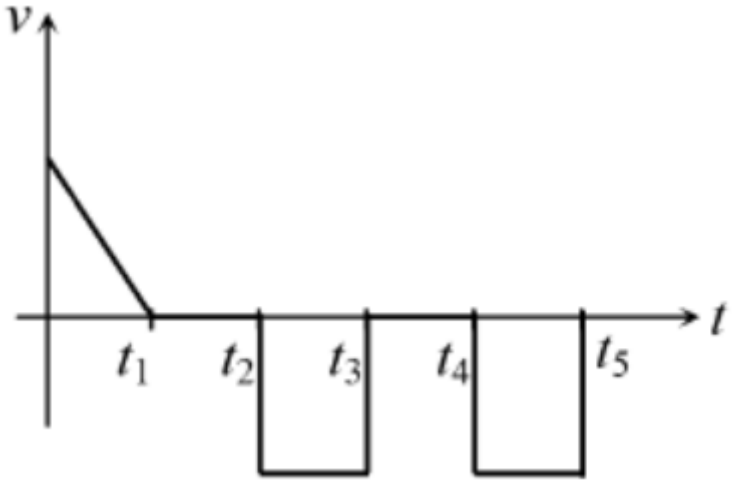
Q.5 Answer:

non-uniform acceleration

Solution:

On differentiating, acceleration $= 0.2\mathrm{t} \Rightarrow \mathrm{a} = \mathrm{f}(\mathrm{t})$

Q.6 Answer:



Solution:

$0 - t_1 \rightarrow$ **uniformly retarded motion**

$t_1 - t_2 \rightarrow$ **particle at rest**

$t_2 - t_3 \rightarrow$ **uniform negative velocity**

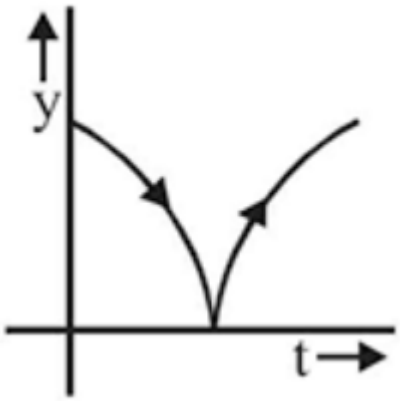
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$t_3 - t_4 \rightarrow$ **particle at rest**

$t_4 - t_5 \rightarrow$ **uniform negative velocity**

$\therefore$

Q.7 Answer:



Solution:

When a ball is dropped on a floor,

$$\mathrm{y} = \frac{1}{2}\mathrm{gt}^2$$

So the graph between y **and** t **is a parabola. Here as time increases, y decreases. When the ball bounces back, then**

$$\mathrm{y} = \mathrm{ut} + \frac{1}{2}\mathrm{gt}^2$$

The graph between y **and** t **will be a parabola. But here as time increases,** y **also increases. So (b) represents the graph.**

Q.8 Answer:

$\frac{1}{3}$

Solution:

During OA, acceleration $= \tan 30^\circ = \frac{1}{\sqrt{3}}\ \mathrm{m/s^2}$

During AB**, acceleration** $= -\tan 60^\circ = -\sqrt{3}\ \mathrm{m/s^2}$

Required ratio $= \frac{1/\sqrt{3}}{\sqrt{3}} = \frac{1}{3}$

Q.9 Answer:

$1/3$

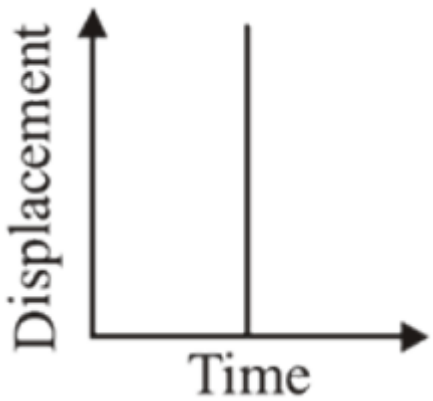
Solution:

$\mathrm{v_A} = \tan 30^\circ$ **and** $\mathrm{v_B} = \tan 60^\circ$

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$\therefore \frac{v_A}{v_B} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1/\sqrt{3}}{\sqrt{3}} = \frac{1}{3}$

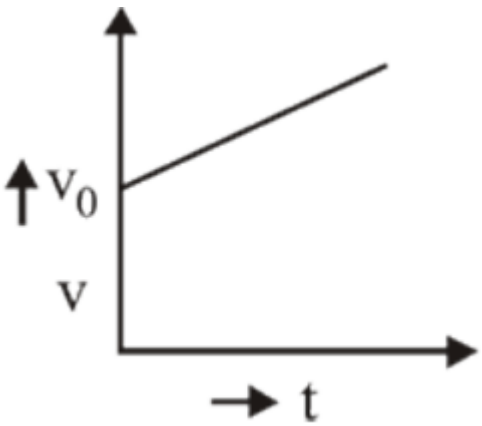
Q.10 Answer:



Solution:

A body moving in a same direction with constant speed is said to be in uniform motion. It can't travel infinite distance instantaneously.

Q.11 Answer:



Solution:

given $x = V_0 t + \frac{1}{2} a t^2$

$\frac{dx}{dt} = V_0 + \frac{1}{2} \times a \times 2t = V_0 + at$

$\Rightarrow V = V_0 + at$

so, option

(c) represents the correct graph.

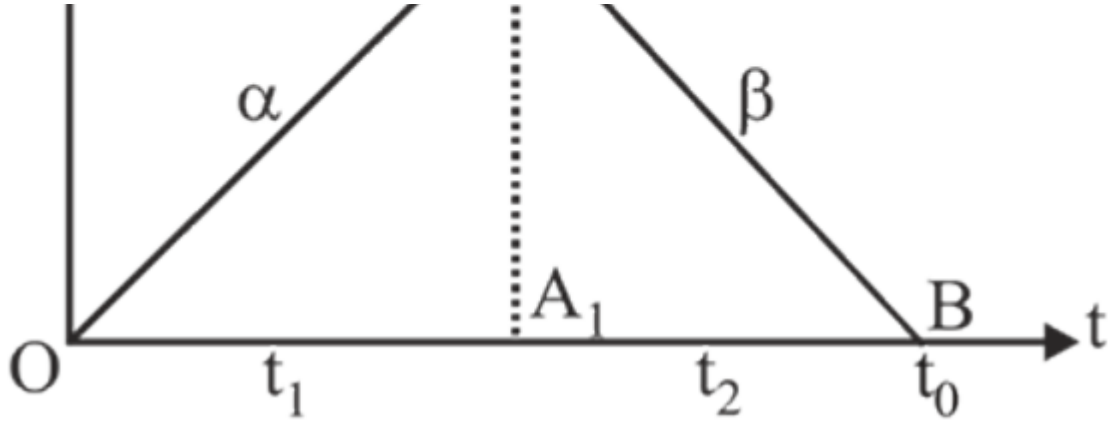
Q.12 Answer:

$\frac{\alpha\beta t}{\alpha+\beta}$

Solution:



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In fig., $\mathrm{AA_1 = v_{max} = \alpha t_1 = \beta t_2}$
But $\mathrm{t = t_1 + t_2 = \frac{v_{max}}{\alpha} + \frac{v_{max}}{\beta}}$
$\mathrm{= v_{max}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = v_{max}\left(\frac{\alpha+\beta}{\alpha\beta}\right)}$
or, $v_{\max} = t\left(\frac{\alpha\beta}{\alpha+\beta}\right)$

Q.13 Answer:

the motion is retarded and finally the particle stops

Solution:

From displacement-time graph, it is clear that in equal intervals of time displacements are not equal infact, decreases and after 40 **s displacement constant i.e. the particle stops.**

Q.14 Answer:

0.5s < t < 1 s

Solution:

Acceleration of the particle $a = \frac{d\vec{v}}{dt} - 2t - 1$
The particle retards when acceleration is opposite to velocity. Hence, acceleration vector and velocity vector should be opposite to each ofher or the dot product of $\vec{a}$ **and** $\vec{v}$ **should be negative.**
$\Rightarrow \vec{a} \cdot \vec{v} < 0$
$\Rightarrow (2t-1)\left(t^2 - t\right) < 0$
$\Rightarrow t(2r-1)(t-1) < 0$
is always positive. $(2t-1)(t-1) < 0$
$\therefore$ **Either** $2t - 1 < 0$ **or** $t - 1 > 0$
$\Rightarrow \quad , < \frac{1}{2}$ **s and** $t > 1$ **s. This is not possible.**
or $2t - 1 > 0$ **and** $t - 1 < 0 \Rightarrow t > \frac{1}{2}$ **s and** $t < 1$ **s.**
Hence, the required time interval is $\frac{1}{2}$ **s** $< t < 1$ **s.**

Q.15 Answer:

2.5 s, 2 m, 0.5 m

Solution:

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(a) Here $u = 5\ \text{m s}^{-1}, a = -2\ \text{m s}^{-2}, v = 0, t = ?$

Using $v = u + at$, **where** $0 = 5 - 2t \Rightarrow t = 2.5\ \text{s}$

$u = 5\ \text{m s}^{-1}, a = -2\ \text{m s}^{-2}, n \leq 2$

(b) Here Using $x_n = u + \frac{a}{2}(2n - 1) = 5 - \frac{2}{2}[2(2) - 1)] = 2\ \text{m}$

(c) Here, if we use the above formula, we will get $x_n = 0$, **but in reality it is not zero. This formula is not applicable for the third second because velocity becomes zero in the third second, i.e., at** $t = 2.5\ \text{s}$. **The particle has a turning point at** $t = 2.5\ \text{s}$. **We have to indirectly calculate the distance travelled in this particular second. That is, we have to determine the distance travelled between** $2\ \text{s} \leq t \leq 2.5\ \text{s}$ **and** $2.5\ \text{s} \leq t \leq 3\ \text{s}$, **and then add the two.**

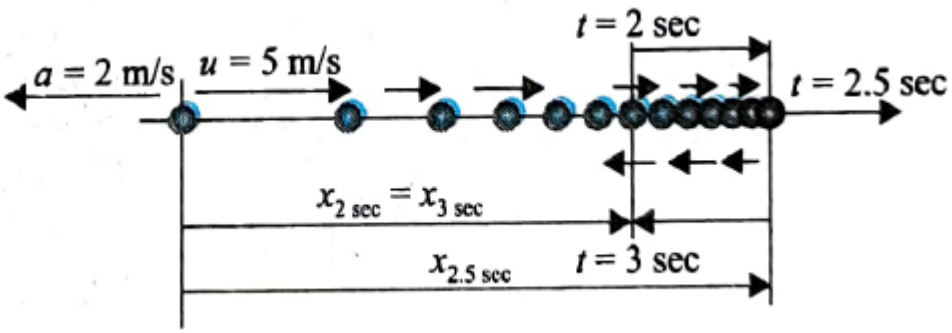
Displacement of the particle at $t = 2.5\ \text{s}$ **is**

$x_{2.5} = \frac{u^2}{2a} = \frac{(5)^2}{2(2)} = 6.25\ \text{m}$

Due to symmetry, the displacement of the particle at $t = 2\ \text{s}$ **and** $t = 3\ \text{s}$ **are same, i.e.,** $x_3 = x_2 = 5(2) + \frac{1}{2}(-2)(2)^2 = 6$

Thus, the distance travelled in the third second is

$$x = (x_{2.5} - x_2) + (x_{2.5} - x_3)$$
$$= (6.25 - 6) + (6.25 - 6) = 0.5\ \text{m}$$

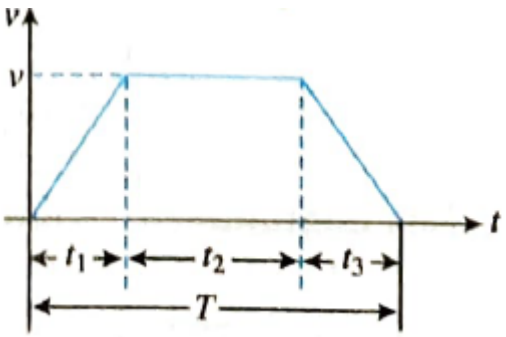


Q.16 Answer:

$\frac{s}{v} + \frac{v}{2}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

Solution:

Let the car accelerate for a time t_1, **move uniformly for a time** t_2 **and then retard for a time** t_3 **as shown in the** $v - t$ **graph.**



Then the total time (T) **of motion of the car is**

$T = t_1 + t_2 + t_3$ **...(i)**

The slope of $v - t$ **graph gives** $\frac{v}{t} = \alpha$ **and** $\frac{-v}{t_3} = -\beta$

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Then $t_1 + t_3 = \frac{v}{\alpha} + \frac{v}{\beta}$

...(ii)

Area under $v - t$ graph gives the total displacement $s =$ Area of the trapezium $= \frac{v}{2}(t_2 + T)$ This gives $t_2 = \frac{2s}{v} - T$ ...(iii)

Substituting $t_1 + t_3$ from Eq. (ii) and t_2 from Eq. (iii) in Eq.

(i) we have $T = \frac{v}{\alpha} + \frac{u}{\beta} - \frac{2s}{v} - T$

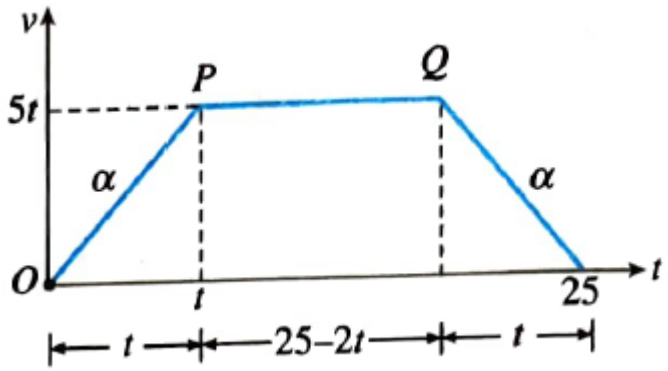
This yields $T = \frac{s}{v} + \frac{v}{2}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

Q.17 Answer:

15s

Solution:

Let t be the time up to which the car accelerates or decelerates. Th maximum velocity attained in this duration is $5t$. The time upto which car moves uniformly $= 25 - 2t$. The velocity-time graph of the motion of car is drawn as shown in Figure.



Given the average velocity is whole time of motion

$v_{\text{av}} = \frac{72 \times 5}{18} = 20 \text{ m s}^{-1}$

The average velocity from the graph can be obtained as

$$v_{\text{av}} = \frac{\text{Total displacement}}{\text{Total time}} = \frac{\text{Area of } \vec{v}-t \text{ graph}}{\text{Total time}}$$

$$20 = \frac{\frac{1}{2} \times [25 + (25 - 2t)] \times 5t}{25} = \frac{\frac{1}{2} \times [50 - 2t)] \times 5t}{25}$$

$$200 = 50t - 2t^2$$

$$\Rightarrow \quad t^2 - 25t + 100 = 0$$

$$\Rightarrow \quad (t - 20)(t - 5) = 0 \Rightarrow t = 5 \text{ s or } 20 \text{ s}$$

But $t = 20$ s is not possible. Hence, $t = 5$ s The time up to which car moves uniformly

$= 25 - 2t = 25 - 2 \times 5 = 15$ s

Q.18 Answer:

$\frac{v_0}{3}$

Solution:

The average velocity is given as:

$v_{\text{av}} = \frac{s}{t} = \frac{\text{Total displacment}}{\text{Total time}}$

We know $a = v\frac{dv}{ds} ightarrow vdv = ads$

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calculate the displacement by substituting $a = -\alpha\sqrt{v}$ (for eet uscation) in $vdv = ads. vdv = (-\alpha\sqrt{v})ds.ightarrow\sqrt{v}dv = -\alpha ds$

we have when the particle slows down from v_0 to 0 , it covers a distance s.

Hence, $\int_{v_0}^{0} \sqrt{v}dv = -\int_{0}^{s} \alpha ds$

This gives $s = \frac{2v_0^{3/2}}{3\alpha}$

Use of $a = \frac{dv}{dt}$:

will find the time of motion by substituting $a = -\alpha\sqrt{v}$ Now we in $a = \frac{dv}{dt}$, we get

$-\alpha\sqrt{v} = \frac{dv}{dt}ightarrow\frac{dv}{\sqrt{v}} = -\alpha dt$

If the particle takes time t to stop, we have

$\int_{v_0}^{0} \frac{dv}{\sqrt{v}} = -\alpha\int_{0}^{t} dt$

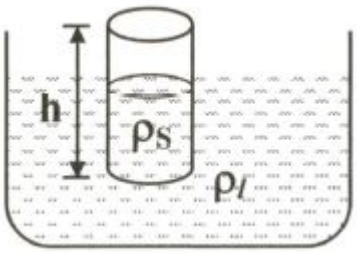
This yields $t = \frac{2\sqrt{v_0}}{\alpha}$

Finally, substituting s and t in $v_{av} = \frac{s}{t}$, we have $v_{av} = \frac{v_0}{3}$

Q.19 Answer:

0.5

Solution:



Equation of motion, $\rho_s \text{Ah}\frac{\text{d}^2\text{x}}{\text{dt}^2} = -x\rho_l Ag$

$\Rightarrow \frac{\text{d}^2\text{x}}{\text{dt}^2} = \frac{\text{g}}{\text{h}}\frac{\rho_t}{\rho_s}\text{x} \quad \Rightarrow \omega = \sqrt{\frac{\text{g}\rho_1}{\text{h}\rho_s}}$

$\Rightarrow \text{a} = \frac{1}{2}; \text{b} = \frac{-1}{2}; \text{c} = \frac{1}{2}; \text{d} = \frac{-1}{2}$

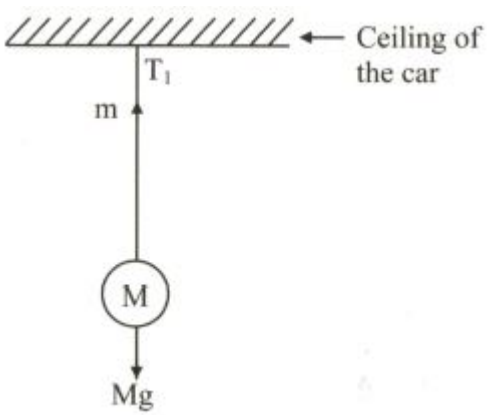
Now $\frac{a^2b}{cd} = \frac{\frac{1}{2}^2\frac{-1}{2}}{\frac{1}{2}\frac{-1}{2}} = \frac{1}{2} = 0.5$

Q.20 Answer:

02.82

Solution:

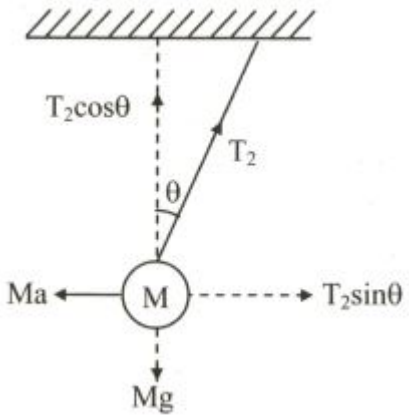
Case I:



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From diagram, $T_1 = mg \therefore \quad v_1 = \sqrt{\frac{T_1}{\mu}} = \sqrt{\frac{mg}{\mu}}$ **....(i)**

Case II:



From diagram, $T_2 \cos\theta = mg, T_2 \sin\theta = ma$

$\therefore \quad T_2^2 \sin^2\theta + \cos^2\theta = m^2g^2 + a^2$

$T_2 = mg^2 + a^2{}^{\frac{1}{2}}$

$v_2 = \sqrt{\frac{T_2}{\mu}} = \sqrt{\frac{m\sqrt{a^2+g^2}}{\mu}} \qquad \ldots . ii$

From (i) and (ii), $\frac{v_1}{v_2} = \sqrt{\frac{g}{g^2+a^2{}^{\frac{1}{2}}}}$

$\Rightarrow \frac{v_2}{v_1} = \frac{g^2+a^2{}^{\frac{1}{4}}}{g^2}$

Taking binomial approximation, we get $\frac{v_2}{v_1} = 1 + \frac{1}{4}\frac{a_2}{g_2} \quad \ldots . \because (1+x)^n \approx 1 + nx$

$\Rightarrow \frac{51}{50} = 1 + \frac{1}{4}\frac{a_2}{g_2}$

$\Rightarrow \frac{a_2}{g_2} = \frac{8}{100}$

$\Rightarrow a = \frac{2\sqrt{2}g}{10}$

$\Rightarrow a = 2.82 \quad ms^{-2}$

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